

3. Scope of Testing

3.1 Overview of Scope

The **Scope of Testing** defines the boundaries of what is included and what is excluded in the ReqRes API Testing Project using Postman. It clearly explains which API functionalities are tested, what type of validations are performed, and what limitations exist in this project.

This helps to ensure that the testing effort is structured, focused, and aligned with real-world API testing practices.

In this project, the scope mainly covers **functional testing of REST APIs**, including request validation, response verification, and execution of different HTTP methods.

3.2 In-Scope Testing Areas

The following areas are included in the scope of this project:

3.2.1 API Functional Testing

The core focus of this project is to validate the functionality of ReqRes REST APIs.

This includes testing of:

- **Login API (Authentication)**
- **Get All Users API**
- **Get Single User API**
- **Create User API**
- **Update User API (PUT request)**
- **Update User API (PATCH request)**
- **Delete User API**

Each API is tested to ensure it performs the expected operation correctly and returns proper responses.

3.2.2 HTTP Methods Validation

The project includes testing of all major HTTP methods used in REST APIs:

- **GET** → Retrieve data (users)
- **POST** → Create new data (user creation, login)
- **PUT** → Full update of existing data

- **PATCH** → Partial update of existing data
- **DELETE** → Remove data

Each method is validated to ensure correct behavior and proper response codes.

3.2.3 Response Validation

The scope includes verifying API responses in detail:

- **Status Code Validation**
 - 200 OK (Success)
 - 201 Created (New resource created)
 - 204 No Content (Delete success)
 - 404 Not Found (Invalid user)
 - **Response Body Validation**
 - Checking JSON structure
 - Verifying presence of required fields (id, name, email, token, etc.)
 - Ensuring correct data is returned
 - **Response Time Validation**
 - Ensuring APIs respond within acceptable time limits
 - Identifying slow or inefficient responses
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3.2.4 Test Case Design and Execution

The project includes:

- Writing manual test scenarios for each API
 - Creating positive test cases (valid inputs)
 - Creating negative test cases (invalid inputs)
 - Executing test cases in Postman
 - Validating actual vs expected results
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3.2.5 Postman Collection Management

The scope includes organizing APIs in Postman:

- Creating a structured collection
- Grouping APIs into folders:
 - Authentication
 - Users
 - Create User
 - Update User
 - Delete User
- Using environments for base URL configuration

- Reusability of requests and variables
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3.2.6 Automation Execution

The project also includes basic automation-level execution:

- Running the collection using **Postman Collection Runner**
 - Executing APIs in sequence
 - Running multiple test cases together
 - Using **Newman CLI** for command-line execution
 - Generating execution reports
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3.2.7 Test Script Validation

Each API includes validation scripts such as:

- Checking status codes
 - Validating response fields
 - Verifying tokens (for login API)
 - Ensuring required data exists in response
 - Basic assertion checks using JavaScript in Postman
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3.3 Out of Scope (Not Included in This Project)

To keep the project focused, the following areas are NOT included:

- No UI (frontend) testing
 - No database validation
 - No performance/load testing (JMeter/Gatling not used)
 - No security penetration testing
 - No real production system testing
 - No authentication token refresh mechanism testing
 - No advanced CI/CD pipeline integration (only basic Newman execution)
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3.4 Assumptions

The following assumptions are made during testing:

- ReqRes API is always available and stable
- API responses are consistent as per documentation
- No backend changes are made during testing

- Internet connection is stable during execution
 - Test data provided by ReqRes is valid and reusable
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3.5 Constraints / Limitations

Some limitations of this project include:

- ReqRes is a **mock API**, not a real production system
 - Limited data sets available for testing
 - No real database connection validation
 - No authentication complexity (simple token-based response)
 - Limited business logic compared to real-world applications
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3.6 Summary of Scope

The scope of this project is focused on **functional API testing of ReqRes REST services using Postman and Newman**. It includes validating endpoints, HTTP methods, responses, and test execution workflows.

This ensures that the APIs behave correctly, return expected results, and meet basic quality standards required in real-world software applications.
